MILITARY SPECIFICATION

PAPER, SHOT SHELL

(For Use In Ammunition)

This specification is approved for use by the U.S. Army Armament Research and Development Command, and is available for use by all Departments and Agencies of the Department of Defense

1. SCOPE

1.1 Scope. This specification covers requirements, quality assurance provisions, packaging for shot shell paper for use in the following ignition cartridges: M2A2, M66A1, M285 and XM299 (see 6.5).

2. APPLICABLE DOCUMENTS

2.1 Issues of documents. The following documents, of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

SPECIFICATIONS

FEDERAL

MMM-A-260	-	Adhesive, Water Resistant (for Sealing Waterproofed Paper)
PPP-B-601	-	Boxes, Wood, Cleated-Plywood
PPP-B-636	-	Boxes, Shipping, Fiberboard

FSC 1395

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, US Army Armament Research and Development Command, Attn. DRDAR-QA, Dover, New Jersey 07801 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T02:45Z
Check the source to verify that this is the current version before use.

PPP-B-1055 - Barrier Material, Waterproofed, Flexible
PPP-D-723 - Drums, Fiber
PPP-T-45 - Tape, Gummed Paper Reinforced and Plain for Sealing and Securing
PPP-T-60 - Tape, Packaging, Waterproof
PPP-T-76 - Tape, Packaging, Paper, (For Carton Sealing)

MILITARY

MIL-L-10547 - Liners, Case and Sheet, Overwrap; Water-Vaporproof or Waterproof, Flexible
MIL-A-48078 - Ammunition Standard Quality Assurance Provisions, General Specification For

STANDARDS

FEDERAL

FED-STD-101 - Preservation, Packaging and Packing Materials; Test Procedures
FED-STD-595 - Federal Standard, Color

MILITARY

MIL-STD-129 - Marking for Shipment and Storage
MIL-STD-147 - Palletized and Containerized Unit Load 40 inch x 80 inch Pallet

(Copies of specifications, standards, and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

TECHNICAL ASSOCIATION OF THE PULP AND PAPER INDUSTRY

(TAPPI) Publications

T 402 Conditioning Paper and Paperboard for Testing
T 403 Bursting Strength of Paper
T 404 Tensile Breaking Strength of Paper and Paperboard
T 410 Basis Weight of Paper and Paperboard
T 411 Thickness (Caliper) of Paper and Paperboard
T 412 Moisture in Paper
T 452 Brightness of Paper and Paperboard
T 450 Air Resistance of Paper

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T02:45Z
Check the source to verify that this is the current version before use.

(Application for copies should be addressed to the Technical Association of the Pulp and Paper Industry, 1 Dunwoody Park, Atlanta, GA 30341.)

UNIFORM CLASSIFICATION COMMITTEE, AGENT

Uniform Freight Classification

(Application for copies should be addressed to the Uniform Classification Committee, Room 1106, 222 South Riverside Plaza, Chicago, IL 60606.)

NATIONAL MOTOR FREIGHT TRAFFIC ASSOCIATION, INC., AGENT

National Motor Freight Classification

(Aplication for copies should be addressed to the American Trucking Associations, Inc., Traffic Department, 1616 P Street, N.W., Washington, DC 20036.)

(Technical society and technical association specifications and standards are generally available from reference libraries. They are also distributed among technical groups and using Federal agencies.)

3. REQUIREMENTS

3.1 Material

3.1.1 Composition. The paper shall be made from 100 percent Northern Softwood semi-bleached kraft with a GE Brightness typically 60-75 when tested as specified in Table VII.

3.1.2 Manufacture. The paper shall be manufactured on a cylinder machine, made into three or four plies with one outer ply colored and with a machine finish.

3.1.3 Color. The color of one side of the papers (wire side) shall be yellow conforming to Yellow Color No. 23655 Table V of FED-STD-595 and the other side shall be semi-bleached kraft felt.

3.2 Physical properties. The finished paper shall conform to the applicable physical properties specified in Table I when tested as specified in Table VII.

Table I, Physical Properties

Characteristics	Requirement
Basis Weight (pounds 24 x 36-500)	68 ± 4
Caliper (Mil)	5.2 ± 0.3

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T02:45Z
Check the source to verify that this is the current version before use.

Densometer (Oil) (Sec./100 ml)	300. ± 100
Mullen Burst (PSI) Minimum	70
Tensile MD/CD (lb/in.) Minimum	75/20
Moisture, percent	4.75 ± 1.25
Water Drop Penetration (Time to absorb 0.05 ml Drop from surface of Paper (Sec.) W/F Sides) each side.	30-70

3.3 Size and roll construction. The paper shall be furnished in rolls of the length and width specified (see 6.1). Rolls shall be wound uniformly and smoothly on non-returnable cores of 3-inch (76 mm) inside diameter with a plus 1/8-inch (3 mm) tolerance. The core length shall be not less than the specified roll width nor greater by more than 1/4-inch (6 mm). Cores shall be fitted with a plug at each end having a 1-inch (25 mm) diameter center hole having plus 1/4-inch (6 mm) tolerance. Rolls shall be suitably restrained to prevent unwinding.

3.3.1 Roll tolerance.

3.3.1.1 Width. A plus 1/8-inch (3 mm) and a minus 1/16-inch 2 mm) tolerance shall be permitted in the cross direction and a plus or minus 1/8-inch (3 mm) tolerance shall be allowed in the machine direction or in both directions.

3.3.1.2 Length. The length of any individual roll shall be not less than 95 percent of the specified length. The average length of rolls in any one lot shall not be less than the specified length.

3.3.2 Splices. Splices shall cover the entire roll width. Splices shall not separate during unwinding of the roll. No roll shall have more than one splice in any 50 yard (46 m) length of the material. Each splice shall be tagged with a colored marker.

3.4 Identification. The paper shall be marked by printing the applicable specification symbols, MIL-P-63381, the year and month of manufacture at least once in each 3 feet (914 mm) of length and at least once in each 18 inches (47 mm) of roll width.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T02:45Z
Check the source to verify that this is the current version before use.

3.5 Workmanship. The paper shall be clean and shall be free from wrinkles, folds, tears, holes, cuts, abrasions, foreign matter, oil spots, surface dirt and dust. Edges shall be clean cut.

3.6 First article inspection. This specification contains provisions for first article inspection. Requirements for the submission of first article samples by the contractor shall be as specified in the contract.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection and standard quality assurance provisions. Unless otherwise specified herein or in the contract, the provisions of MIL-A-48078 shall apply and are hereby made a part of this detail specification.

4.2 Classification of inspections. The following types of inspections shall be conducted on this item:

a. First Article Inspection (see 4.3)
b. Quality Conformance Inspection (see 4.4)

4.3 First article inspection.

4.3.1 Submission. The contractor shall submit a first article sample as designated by the Contracting Officer for evaluation in accordance with provisions of 4.3.2. The first article sample shall consist of two (2) rolls of the paper.

4.3.2 Inspection to be performed. See MIL-A-48078 and Table VII specified herein.

4.3.3 Rejection. See MIL-A-48078.

4.4 Quality conformance inspection.

4.4.1 Inspection lot formation. Inspection lots shall comply with lot formation provisions of MIL-A-48078.

4.4.2 Examination - See MIL-A-48078

a. Sampling plans. Unless otherwise specified in this specification sampling for inspection shall be in accordance with MIL-STD-105. If any sample fails to meet any test requirement, the lot represented by the sample shall be rejected. The classification of Defects for Composition (see 3.1.1) and Physical Properties (see 3.2) shall be Major.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T02:45Z
Check the source to verify that this is the current version before use.

b. End item inspection. The end item shall be examined for the defects as set forth below at the inspection levels and acceptable quality levels (AQL's) set forth in 4.4.2.7. Random samples shall be drawn from each lot of the same type and size for examination of visual, dimensional, and packaging defects. The lot size shall be all the rolls of the same type submitted at one time and shall be expressed in finished rolls for examinations in 4.4.2.1 thru 4.4.2.4, in units of shipping containers for examinations in 4.4.2.5 and in palletized unit loads, for examination in 4.4.2.6.

4.4.2.1 Visual examination. The sample unit for this examination shall be three randomly selected samples of two yards in length and the full roll width. No sample units shall be taken from the first or last convolutions of the roll. Both sides of the sheet shall be examined. Defects of each type shall be scored only once within a sample unit.

TABLE II. Classification of visual defects (Minor Defect)

Examine	Defect
Workmanship	Not clean; presence of any foreign matter, dirt, sand, grit, or oil spots. Presence of any cut, tear puncture, chafed spot, wrinkle or fold. Edges not clean cut, ragged, crushed or uneven.
Identification markings	Missing, illegible, incorrect, incomplete, do not conform to legend specified in 3.4.
Color	Visual, Yellow Color No. 23655 Tavle V of FED-STD-595 and the other side shall be semi-bleached kraft felt.

4.4.2.2 Examination for dimensional defects. The sample unit for this examination shall be one roll. The sample shall be examined for the defects listed in Table III.

TABLE III. Classification of dimensional defects (Minor Defect)

Examine	Defect
Width	Varies by more than + 1/8 inch (3 mm) and -1/16 inch (2 mm) in cross direction Varies by more than ± 1/8 inch (3 mm) in machine direction or in both directions.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T02:45Z
Check the source to verify that this is the current version before use.

Core plugs	Plug holes not centered. Diameter of plug hole less than 1 inch (25 mm) or more than 1-1/4 inches (32 mm).
Core	Length less than specified width of roll material or greater by more than 1/4 inch (6 mm). Inside diameter less than 3 inches (76 mm) or more than 3-1/8 inches (79 mm).
Identification markings	More than 36 inch (914 mm) distance along length between groups of marking. More than 18 inches (457 mm) in width between groups of markings.

4.4.2.3 Examination for defects in roll construction. The sample unit for this examination shall be one roll. The sample shall be inspected for the defects in Table IV.

TABLE IV. Classification of roll construction defects (Minor Defect)

Examine	Defect
Assembly of roll	Not suitably restrained to prevent unwinding. Material not wound evenly and tightly on roll causing wrinkles, creases or telescoping. Material not wound on core. Core broken, collasped, crushed or mutilated. Core plugs omitted or not as specified.
Unwinding of rolls (examine both sides)	When unwound, material sticks together to the extent that unrolling causes tearing or injury to any surface. Roll not continuous. More than 1 splice in any 50 consecutive yards (46 m) of roll. Splice not evenly and neatly made; does not cover entire width of material. Splice comes apart during unwinding Splice not flagged with colored marker.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T02:45Z
Check the source to verify that this is the current version before use.

4.4.2.4 Examination for length of roll. The sample unit for this examination shall be one roll. The length of the roll examined shall be not less than 95 percent of the specified length (Minor defect).

4,4,2,5 Inspection of packaging. An inspection shall be made to determine that packing and markings comply with requirements of Section 5. The sample unit for this inspection shall be one shipping container, fully packed except it need not be closed. Closed shipping containers, fully prepared for delivery, shall be examined for closure defects.

TABLE V. Inspection of packing and marking (Minor Defect)

Examine	Defect
Packing	Container not as specified. Not in accordance with contract requirements. Not level specified. Rolls not overwrapped as specified. Wrapper material other than specified for applicable level. Headers, inside or out, omitted or not of same material as overwrap Closures not accomplished by specified methods or materials. Inadequate application of components such as: incomplete closure of case liner, container flaps, loose or inadequate sealing, strapping or stapling. Bulged or distorted container.
Markings	Illegible, incomplete, omitted, or not in accordance with contract requirements.

4.4.2.6 Examination for palletization. An examination shall be made to determine that the palletization complies with the Section 5 requirements. Defects shall be in accordance with Table VI. The sample unit shall be one palletized unit load fully prepared for delivery. The lot shall be the number of palletized unit loads in the end item inspection lot. The inspection level shall be S-1 and the acceptable quality level (AQL) shall be 6.5 defects per hundred units.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T02:45Z
Check the source to verify that this is the current version before use.

TABLE VI. Examination of palletization (Minor Defect)

Examine	Defect
Finished dimension	Length, width or height exceeds specified maximum requirement.
Palletization	Not as specified. Pallet pattern not as specified. Wood cap not used over and under the load. Load not bonded with required straps as specified.
Weight	Exceeds maximum load limits.
Marking	Omitted; incorrect; incomplete; illegible; of improper size, location, sequence or method of application.

4.4.2.7 Inspection levels and acceptable quality levels (AQL's) for examinations. The inspection level for determining the sample size shall be S-3 and the acceptable quality levels (AQL's) shall be 4.0 expressed in terms of defects per hundred units. The same randomly selected rolls shall be used for the examinations in 4.4.2.1 thru 4.4.2.4.

4.4.3 Testing All tests shall be conducted on material that was conditioned in an atmosphere in accordance with method T 402 of TAPPI. In addition, the testing atmosphere shall be the same as the conditioning atmosphere.

4.4.3.1 End item testing. The end item shall be tested for the applicable characteristics listed in this specification. The lot size shall be the number of rolls in the end item inspection lot and the sample unit shall be one full roll-width sheet of 48 inch (1.2 m) length. The sample size shall be selected in accordance with the following sampling plan. Test results shall include all values on which the results are based and there shall be no failures to meet the sample unit requirements.

Lot size (rolls)	Sample size (number of sample units)
800 or less	2
801 up to and including 22,000	3
22,001 or more	5

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T02:45Z
Check the source to verify that this is the current version before use.

TESTING METHODS AND SAMPLES FOR PHYSICAL CHARACTERISTICS

TABLE VII

CHARACTERISTIC	TEST METHOD	NUMBER DETERMINATION PER SAMPLE UNIT
Atmospheric conditions (See 4.4.3)	T 402	
Basic weight (24 x 36-500)	T 410	10
Caliper	T 411	10
Tensile		
Machine direction	T 404	10
Cross direction	T 404	10
Brightness of paper	T 452	7
Moisture	T 412	1
Mullen Burst	T 403	20
Densometer	T 460	10
Water Drop Penetration	4.4.1	5

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T02:45Z
Check the source to verify that this is the current version before use.

4.4.4 Inspection equipment. The government reserves the right to inspect the contractor's equipment and determine that he has available and utilizes correctly, measuring and test equipment of the required accuracy and precision and that the instruments are of the proper type and range to make measurements of the required accuracy. Commercial inspection equipment shall be employed where applicable for all tests and examinations specified in 4.5. The contractor is responsible for assuring that proper calibration procedures are followed. Government approval of all inspection equipment is required prior to its use for acceptance purposes (see Section 6 of MIL-A-48078 and 6.2 herein).

4.5 Test Procedures (see 4.4.3)

4.5.1 Water Drop Penetration. Five specimens of paper shall be cut into 3 inches squares. A specimen shall be placed under the tip of a 10 ml buret which is positioned at a 45° angle supported by a buret stand and clamps. The buret shall be lifted slightly to allow one drop of water to form on the tip of the buret. The drop of water shall be carefully transferred to the paper. The laboratory timer shall be started as the drop touches the surface of the paper. The timer shall be stopped when the water is completely absorbed. This point shall be judged by observing the reflected light from the droplet at an obtique angle. When the shiny appearance of the wet surface disappears, the absorption shall be considered complete. The timer shall be stopped and the time recorded. The average of five tests absorption time of each side shall be reported.

5. PACKAGING

5.1 Packing. Packing shall be level A, B or Commercial as specified (see 6.1).

5.1.1 Level A. Paper in roll form as specified in 3.4, shall be packed either in a snug-fitting fiber drum conforming to type III, grade D of PPP-D-723; fiberboard shipping container conforming to type RSC, type SF, grade VSs of PPP-B-636; or wrapped overall with waterproof barrier material conforming to PPP-B-1055 (except if class C-1 is used, two thicknesses shall be required). The barrier material wrapper shall be closed at the ends by means of inside and outside headers made of the same material as the wrapper. Seams and headers shall be sealed with an adhesive conforming to type I or II, grade B, class 2 or 3 of MMM-A-260; or sealed with 2-inch (51 mm) minimum width tape conforming to either type III, class 1 of PPP-T-60 or PPP-T-76. The adhesive or tape shall be

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T02:45Z
Check the source to verify that this is the current version before use.

applied to the barrier material wrapper and headers in a manner which will effect a watertight seal. Each fiber drum shall be closed with pressure-sensitive tape conforming to either type III, class 1 of PPP-T-60, or PPP-T-76 in accordance with the applicable requirements of PPP-D-723. Each fiberboard shipping container shall be closed, waterproofed and reinforced in accordance with the appendix of PPP-B-636. When specified (see 6.1), a multiple quantity of paper, shall be packed in a snug-fitting shipping container conforming to overseas type, style A, type 2 load of PPP-B-601. Each shipping container shall be waterproofed with a sealed case liner conforming to type I or II, grade C of MIL-L-10547. Each container shall be closed and reinforced in accordance with the appendix of PPP-B-601.

5.1.2 Level B. The paper in roll form shall be packed in a snug-fitting fiber drum conforming to type I, grade D of PPP-D-723; fiberboard container conforming to style RSC, type CF (variety SW) or SF, class domestic, grade 275 of PPP-B-636; or wrapped overall with 80-pound (130 g/m^2) minimum basis weight (24 by 36-500) heavy duty wrapping paper. Each paper wrapped roll shall be closed at the ends by means of inside and outside headers made of the same material as the wrapper. Seams and headers shall be securely sealed with an adhesive commercially used for this purpose or by a 2-inch (51 mm) minimum width gummed paper tape conforming to type III, grade C or PPP-T-45. Each fiber drum shall be closed in accordance with the applicable requirements of PPP-D-723. Each fiberboard container shall be closed in accordance with method II as specified in the appendix of the container specification.

5.1.2.1 Weather-resistant shipping container. When specified (see 6.1), the fiberboard shipping container shall be grade V3c, V3s, or V4s fiberboard box fabricated in accordance with PPP-B-636 and closed in accordance with the appendix of the box specification.

5.1.3 Commercial. The paper in roll form shall be packed in a manner to insure carrier acceptance and safe delivery to destination at the lowest transportation rate for such supplies. Containers and packing shall comply with Uniform Freight Classification and National Motor Freight Classification.

5.2 Palletization. When specified (see 6.1), The paper packed as specified in 5.1 shall be palletized in accordance with load type XIII of MIL-STD-147.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T02:45Z
Check the source to verify that this is the current version before use.

Rolls of paper shall be stacked vertically in a single course with a wood cap over and under the load. A strap shall be positioned around the load outside of and at the vertical enter of the rolls of the paper and shall be the first strap applied to the load. Each prepared load shall be bonded with primary and secondary straps in accordance with bonding means K and L.

5.3 Marking. In addition to any special marking required by the contract or order, shipping containers and wrapped rolls (as applicable) shall be marked in accordance with MIL-STD-129.

6. NOTES

6.1 Ordering data. Procurement documents should specify the following: (see MIL-A-48078)

a. Title, number, and date of this specification.
b. Width and length per roll required (see 3.3)
c. Selection of applicable level of packing (see 5.1).
d. When weather resistant fiberboard shipping containers are required for level B packing (see 5.1.2.1).
e. Where wood shipping containers are required for level A packing (see 5.1.1).
f. When palletization is required (see 5.2).

6.2 Submission of inspection equipment designs for approval. See MIL-A-48078. Submit designs for as required to Commander, US Army Armament Reserch and Development Command, ATTN: DRDAR-QAR-R, Dover, New Jersey 07801.

6.3 Prior approval of the Contracting Officer is required for use of equivalent test methods. A description of the proposed method should be submitted through the Contracting Officer to: Commander, US Army Armament Research and Development Command, ATTN: DRDAR-QAR-R, Dover, New Jersey 07801. This description should include but not be limited to the procedures used, the accuracy and precision of the method, test data to demonstrate the accuracy and precision and drawings of any special equipment required.

6.4 Unless otherwise specified, certificates of compliance shall be submitted for the paper.

6.5 The material described in this specification was previously described in a Source Control Drawing 9243255 which was prepared by Picatinny Arsenal.

Custodian:
Army-AR

Preparing Activity
Army-AR
Project NO. 1395-A230

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T02:45Z
Check the source to verify that this is the current version before use.

STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

INSTRUCTIONS: This form is provided to solicit beneficial comments which may improve this document and enhance its use. DoD contractors, government activities, manufacturers, vendors, or other prospective users of the document are invited to submit comments to the government. Fold on lines on reverse side, staple in corner, and send to preparing activity. Attach any pertinent data which may be of use in improving this document. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity. A response will be provided to the submitter, when name and address is provided, within 30 days indicating that the 1426 was received and when any appropriate action on it will be completed.
NOTE: This form shall not be used to submit requests for waivers, deviations or clarification of specification requirements on current contracts. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements.

DOCUMENT IDENTIFIER *(Number)* AND TITLE
MIL-P-63381 Paper, Shot Shell (for use in Ammunition)

NAME OF ORGANIZATION AND ADDRESS OF SUBMITTER

☐ VENDOR ☐ USER ☐ MANUFACTURER

1. ☐ HAS ANY PART OF THE DOCUMENT CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE? ☐ IS ANY PART OF IT TOO RIGID, RESTRICTIVE, LOOSE OR AMBIGUOUS? PLEASE EXPLAIN BELOW.

A. GIVE PARAGRAPH NUMBER AND WORDING

B. RECOMMENDED WORDING CHANGE

C. REASON FOR RECOMMENDED CHANGE(S)

2. REMARKS

SUBMITTED BY *(Printed or typed name and address — Optional)*

TELEPHONE NO.

DATE

DD FORM 1 OCT 76 1426 EDITION OF 1 JAN 72 WILL BE USED UNTIL EXHAUSTED.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T02:45Z
Check the source to verify that this is the current version before use.